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Review Article

Is the metaverse really coming to fruition? A survey of applied metaverse and extended reality

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ABSTRACT

This survey examines the current state of the Metaverse, encompassing its fundamental concepts, technological framework, practical applications, and user experience to evaluate its stage of development. This paper reviews the core concepts of the Metaverse and Extended Reality (XR) and evaluates the latest advancements in hardware and software technologies. Furthermore, it examines the Metaverse's typical applications in four key domains: education, training, medicine, and mixed life, while summarizing user feedback to identify its advantages and challenges. The feedback indicates that the Metaverse offers notable benefits, including immersive experiences, enhanced training effectiveness, cost efficiency, and improved safety. However, significant challenges remain, such as hardware performance limitations, software inefficiencies, user discomfort, health risks, and social and ethical concerns. Our analysis suggests that while the Metaverse has yet to reach full maturity, it holds great potential for future development. To further advance the field, this paper highlights key research priorities in artificial intelligence(AI), quantum computing, and social governance, providing insights for future studies.

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1. Introduction

Metaverse, a persistent and immersive environment that parallels or integrates with the real world, has garnered significant attention from both academia and industry in recent years. The rebranding of Facebook to "Meta" in 2021 served as a catalyst for a global surge in Metaverse-related research, investments, and technological advancements [1]. However, despite continuous efforts to develop and refine the Metaverse, the field still faces numerous challenges, including hardware limitations, a lack of high-quality content, immature interaction methods, and ethical and societal concerns. These obstacles raise critical questions regarding the true realization of the Metaverse and its future trajectory. This survey centers on the Metaverse, providing an in-depth investigation of its technological foundations, user applications, and societal challenges.

The concept of the Metaverse is not a recent development. Early 2D online communities and basic digital games laid the foundation for virtual social interactions [2]. As advancements in computer graphics and rendering technologies progressed, 3D virtual environments became mainstream, finding applications in computer-aided design (CAD) [3], immersive gaming [4], and virtual collaboration platforms [5]. These developments significantly enhanced the realism and immersiveness of digital interactions. In recent years, the advent of XR has further blurred the boundaries between virtual and physical spaces. XR encompasses Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), serving as the technological backbone of the Metaverse [6]. VR enables fully immersive digital experiences, while AR overlays virtual elements onto the real world, enhancing user perception. These technological advancements have not only facilitated the realization of the Metaverse concept but have also profoundly transformed how individuals perceive and engage with digital spaces. Notably, the term "Metaverse" was first introduced in the science fiction novel *Snow Crash*, where it referred to a 3D VR-based internet world [7]. The modern Metaverse, however, has evolved beyond this initial vision to encompass interconnected virtual spaces, immersive interaction environments, and a seamless integration of digital and physical ecosystems.

In recent years, academic research on the Metaverse has explored its technological architecture, key innovations, and practical applications. Some studies focus on the Metaverse's fundamental infrastructure, technical composition, and developmental trends, while others examine specific use cases, industry developments, and associated challenges. Early research (pre-2015)

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primarily concentrated on 3D virtual worlds and basic interaction technologies. For instance, Dionisio et al. provided a systematic review of 3D virtual world development and projected future trends for the Metaverse, discussing the evolution of immersive virtual environments [8]. However, these early studies predominantly addressed computer graphics, 3D modeling, and foundational virtual world interactions, with limited exploration of deeper aspects such as human–computer interaction, AI, and integration with real-world social structures. In contrast, more recent research (post-2021) has shifted toward XR, AI, cybersecurity, and societal implications. Wang et al. conducted an extensive analysis of the Metaverse's technical architecture, core applications, and technological challenges, highlighting the role of emerging technologies in shaping the Metaverse [6]. Nevertheless, their study primarily focuses on the current state of technological advancements, with limited discussion on the practical implementation of the Metaverse and the associated user experiences, providing few case studies. Similarly, Rokhsaritalemi et al. examined the development of MR technology and its challenges but concentrated on MR itself without thoroughly investigating how XR and MR integrate into the broader Metaverse ecosystem [9]. Pooyandeh et al. focused on AI-driven security concerns in the Metaverse, covering identity management, data security, and potential cyber threats, while Huang et al. analyzed privacy protection, authentication, and data security risks [10,11]. Although these studies highlight crucial security concerns, they primarily emphasize information security while lacking a systematic assessment of real-world Metaverse applications, user experience challenges, and the intersection of technological and societal issues. Additionally, while some research has addressed ethical, privacy, and legal issues, there remains a gap in understanding the technological frameworks, societal acceptance, and long-term implications of the Metaverse.

Despite the valuable insights offered by prior studies, several research gaps persist. First, most existing studies focus on the theoretical framework of Metaverse technology rather than its practical implementation, resulting in a lack of empirical analysis on real-world Metaverse platforms, applications, and user experiences. Second, there is an insufficient examination of user interactions within the Metaverse, including user adaptability and challenges associated with digital environments. Third, the current body of literature lacks a comprehensive evaluation of the Metaverse's development across various application domains, such as education, healthcare, training, and mixed life (encompassing social interactions, entertainment, and commercial activities), leaving industry trends and technical barriers underexplored.

To address these research gaps, this study employs a systematic evaluation of the current state of the Metaverse through case studies and user feedback analysis. Specifically, this research aims to: (1) define the core concepts and key technologies underlying the Metaverse, such as XR, spatial computing, and Digital Twin, to establish a unified technical framework. (2) assess the current state of Metaverse-related hardware and software, systematically identifying development trends and technological bottlenecks through case analysis. (3) investigate the real-world applications of the Metaverse in education, training, healthcare, and mixed life, incorporating user feedback to evaluate its practical value and limitations. (4) summarize the key technological, societal, and user experience challenges facing the Metaverse while proposing future research directions, including AI-driven content generation, quantum computing for real-time interactions, and advancements in digital governance systems. Unlike previous research, this study not only synthesizes Metaverse technological advancements but also integrates real-world case studies and user feedback to provide a more comprehensive and practice-oriented evaluation, offering valuable insights for future technological optimizations and policy considerations.

The remainder of this paper is organized as follows. Section 2 introduces fundamental concepts related to the Metaverse and XR, establishing the theoretical foundation. Section 3 reviews the technological advancements underpinning the Metaverse, including hardware systems, software architectures, and interactive technologies, while analyzing current development status and technical challenges. Section 4 explores real-world Metaverse applications in education, training, healthcare, and mixed life, summarizing case studies and user feedback to identify prevailing trends and obstacles. Section 5 evaluates the feasibility of the Metaverse, assessing its technological maturity, user experience, and societal acceptance, along with a detailed discussion of both positive and negative perspectives. Section 6 examines the primary challenges facing the Metaverse and proposes future development trajectories, offering insights into its long-term evolution. Section 7 presents potential solutions and future research directions to address existing issues and facilitate the further advancement of the Metaverse. Finally, Section 8 provides a summary of the key challenges and development opportunities for the Metaverse, revisiting the main discussions from previous sections and outlining its future trajectory.

2. Concepts in the metaverse

When we talk about the Metaverse, we probably have many questions. What is the Metaverse? What are Extended, Virtual, Mixed, and Augmented Reality? What is the difference between MR and AR? Why is Spatial Computing very important for XR? What is a Digital Twin? And what is the relationship between the above technologies and the Metaverse? This section will answer these questions by providing an analysis and supporting evidence based on the relevant research.

2.1. Metaverse

The word first appeared in Snow Crash [7] in 1994, and it now has a much broader meaning than when it first appeared. The original Metaverse depicted a 3D virtual world, but the current Metaverse is an environment containing both virtual and real elements.

It can be both a virtual world, a connection of multiple virtual worlds, or even an augmented physical world [6,8,12–15]. The commonality is that it allows people to live through digital virtual avatars. The complexity of the Metaverse is such that a single technology cannot realize it, and it requires the collaboration of multiple technologies. These include XR, Spatial Computing, Digital Twin, Blockchain, AI technologies, and other technologies that can help build the Metaverse, as shown in Fig. 1. We already have some technology to build the Metaverse, and we know what it is, but how can we actually see it? Today's solution is to see the Metaverse through XR technology, so what is XR? The XR will be explored in the next section.

2.2. Extended reality

XR has now become a generalized term describing the entirety of these immersive technologies encompassing all immersive technologies involving VR and the combination of the virtual and the real [6,16–18]. The term XR was selected during the development of immersive technologies. With the gradual development of VR, MR, and AR immersive technologies, a new term was needed to describe them, and the word "Extended Reality" was chosen. One interesting thing here is that XR clearly represents all three immersive technologies, but it could actually be seen as including just VR and MR as well as shown in Fig. 1. This leads us to think more deeply: what does VR mean, what does MR mean, what does AR mean, and what is the relationship between MR and AR?

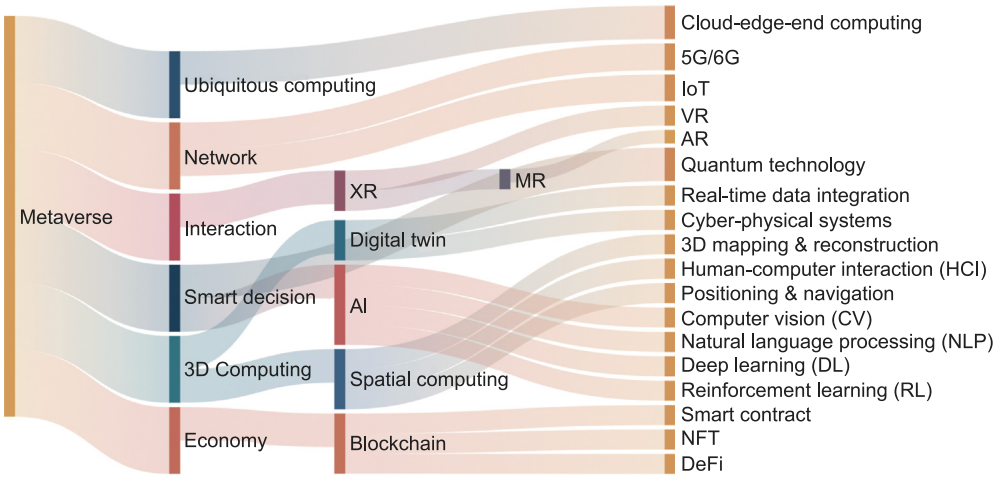

Fig. 1. Technical components of the metaverse.

Fig. 2. The game scene of Beat Saber [21].

Fig. 3. Utilizing MR in work [23].

2.3. Virtual reality

VR has a long history, and its initial development can be traced to the 1960s, first proposed by computer scientist Ivan Sutherland [19,20]. VR represents a technology that creates an entirely virtual 3D environment, and it allows users to immerse and interact in a fully synthetic world. It isolates the user's senses from the real world, and perceptions and interactions occur in this virtual world without the user being able to see or directly interact with the real world. As such, VR provides a completely virtual experience that simulates reality or creates fantastical scenarios. such as the VR game "Beat Saber" scenario shown in Fig. 2. Now we know the other end of the fully real world is the fully virtual world, so what lies between the two endpoints? This is where MR comes into play, and the next section will explore this concept in detail.

2.4. Mixed reality

MR is a technology that combines the real and virtual worlds, allowing users to experience and interact with both simultaneously within the same visual environment [9,19,22]. The concept was first introduced by Paul Milgram and Fumio Kishino in 1994, where they defined MR as any point on a continuum between complete reality (the real world) and complete virtuality (virtual reality) [19].

MR realizes real-time interaction between virtual and physical objects by fusing the virtual and real worlds. Users can not only see virtual elements in the real environment but also interact with these virtual elements and real objects at the same time. The seamless integration of virtual objects with the physical world allows them to become real-time assistants at work, as shown in Fig. 3.

Since MR has both virtual and real contexts, this can't help but make us think more closely about the difference between focusing on the virtual and focusing on the real. Focusing on the virtual can be called Augmented Virtuality, so how about focusing on the real? That is AR, which will be introduced in the next section.

2.5. Augmented reality

AR is a technology that overlays computer-generated information into the real world, enhancing the user's perception of their environment [19,24–26].

Other researchers have added to the definition of interactivity in AR [27,28]. When AR was added to the interactive function, it belonged to MR, focusing on the real side. AR originated in the 1990s when Calder and others coined the term "augmented reality" while developing a virtual guidance system to help Boeing electricians assemble airplane harnesses [24]. And now AR technology has surrounded our lives. It is utilized in various emoticons and short videos on streaming media platforms, just like the bird in Fig. 4, allowing interaction through your phone's screen. We can find that AR actually needs to capture the spatial information to overlay a digital thing well onto the real world. Furthermore, not only does AR need it, but VR and MR also need to capture spatial information to compute the spatial. What is spatial computing? We explain it in the next section.

Fig. 4. AR filters in short videos [29].

Fig. 5. Meta Quest Boundary [35].

2.6. Spatial computing

Spatial computing is a term coined by Simon Greenwold in 2003 [30]. Although it has since been defined differently by other researchers, the common core concept of spatial computing is to enable devices to sense and understand their environment and to convert physical information into digital information. This ability allows XR technology to seamlessly and naturally integrate digital information into physical space, allowing users to interact with and operate with digital content in reality [31–34]. As shown in Fig. 5, spatial computing technology is utilized in a specific VR application to scan a physical space and create a suitable virtual environment.

2.7. Digital Twin

The concept of Digital Twin dates back to 2003 [36], but it was not until 2012 that NASA formalized the term "Digital Twin" [37]. Digital Twin is a very interesting concept that refers to a virtual copy of a physical entity or system. This virtual copy is kept in synchronous with the actual physical object through real-time data exchange and can dynamically simulate its state and behavior. At its core, it enables the performance and behavior of physical systems to be analyzed, predicted, and improved in real-time in a digital space through continuous feedback between virtual models and real objects [38–43]. This technology is not limited to simple replication. It integrates multiple information sources, such as sensor data, historical data, physical models, and machine learning algorithms, to achieve dynamic simulation and optimization of physical entities, and creates a digital aircraft, as shown in Fig. 6. For and with Metaverse, Digital Twin technology can bring real-world objects and environments into Metaverse, thus enhancing the realism and interactivity of the virtual world and making the interaction between the virtual and the real more intimate and intuitive.

Fig. 6. Digital twin of airplane [44].

2.8. Summary

We know what a Metaverse is and what these technical terms involved in a Metaverse actually mean. Now the time has come to explain exactly how they relate to each other. The Metaverse is an umbrella term for a wide range of real and virtual environments, and XR technology is the tool for accessing the virtual world. XR covers VR, MR, and AR, where VR is fully virtual, MR fuses virtual and real, and AR augments reality. Spatial computing underpins these interactions, and Digital Twin simulates reality through virtual models and can be used to augment the reality of the Metaverse interactions. They collaborate and complement each other to build the Metaverse.

3. Modern technology of metaverse and XR

Having established the fundamental technical concepts of the Metaverse, we now turn our focus to the core technologies that enable its realization. At its core, the Metaverse is a digital ecosystem that seamlessly integrates the physical and virtual worlds, facilitating immersive and interactive experiences. In this section, we will explore the hardware and software technologies that drive the Metaverse, examining their roles in bridging the gap between the physical and virtual realms [45–47].

3.1. Hardware

XR hardware can be classified into three primary categories based on its functionality, as shown in Fig. 7.

- **Imaging hardware**, responsible for display and optical processing, directly impacting visual experience and immersion.
- **Core hardware**, which includes processing units and tracking systems that drive real-time computation and spatial awareness.
- **Auxiliary hardware**, which enhances user interaction through additional sensors, controllers, and haptic feedback.

These components collectively handle input, output, and processing, ensuring seamless interaction between the physical and virtual worlds.

3.1.1. Imaging hardware

The performance of imaging hardware in XR devices is primarily determined by three key parameters: field of view (FoV), pixels per degree (PPD), and refresh rate. These factors directly impact visual immersion, image clarity, and user comfort.

FoV determines the extent of the virtual environment visible to the user [47]. A minimum FoV of 90° in all directions aligns

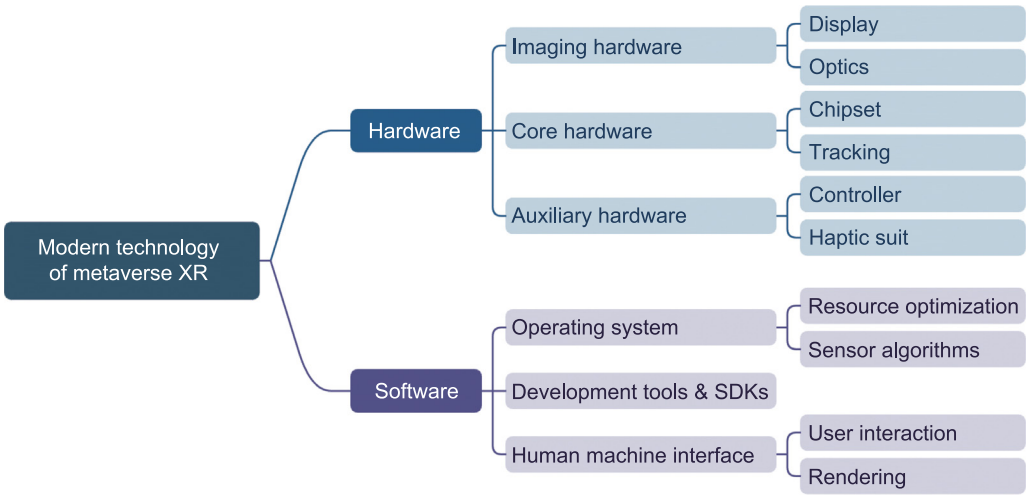

Fig. 7. Technology overview.

with natural human vision and reduces motion sickness [48–50]. A larger FoV enhances immersion by reducing excessive head movement and preventing the "binocular effect", where users feel constrained by artificial tunnel vision.

PPD quantifies the number of pixels displayed per degree of vision. Higher PPD reduces the screen door effect (SDE), where visible gaps between pixels create a mesh-like appearance that disrupts immersion [51]. Modern XR headsets strive to minimize SDE by increasing pixel density and incorporating subpixel rendering techniques and diffusion layers to mitigate visible pixel gaps and improve perceived image continuity.

Refresh Rate influences visual smoothness and user comfort. A minimum of 75 Hz is recommended to reduce motion sickness and prevent visual lag in dynamic environments [52]. Higher refresh rates, such as 90 Hz or 120 Hz, provide smoother motion transitions, improving the overall user experience.

Display technologies. XR devices typically use displays ranging up to 7 inches. The primary display technologies include liquid crystal displays (LCD), organic light-emitting diodes (OLED), mini-LEDs (mLED), and micro-LEDs (μLED), each offering varying levels of brightness, contrast, and power efficiency [53].

LCD use liquid crystals to control light transmission. These displays rely on an LED backlight, which can limit contrast and black levels compared to self-emissive technologies like OLED.

OLED consist of organic compounds that emit light when an electric current is applied. Each pixel is self-emissive, meaning it generates its own light without a backlight. OLEDs offer deeper blacks, superior contrast, and faster response times, making them ideal for VR headsets. However, they are susceptible to burn-in, where prolonged static images cause permanent discoloration.

mLEDs use smaller, densely packed LED backlights to achieve better local dimming and enhanced contrast ratios. While they offer improved brightness and contrast over traditional LCDs, they still depend on a backlight and are not self-emissive.

μLED are self-emissive, with each pixel functioning as an individual LED. They offer higher brightness, improved energy efficiency, and longer lifespan without burn-in risk. Their high contrast ratios and fast response times make them well-suited for XR applications requiring precise color rendering and motion clarity. However, challenges in miniaturizing LEDs while maintaining uniform brightness and reducing manufacturing defects limit their adoption.

Optics in XR devices. Optics focus and direct light from small displays to the user's eyes. Due to the compact nature of XR headsets, human eyes cannot naturally focus on screens positioned so close. Therefore, optics are required to reshape and project the image into a viewable form. However, this process introduces challenges, such as optical efficiency loss and image distortion, which need to be addressed through advanced lens technologies.

The most common optics in VR devices include Fresnel lenses, pancake lenses, and aspherical lenses.

Fresnel Lenses use a series of concentric, ring-shaped segments to bend light, similar to a traditional convex lens but in a thinner, lighter form. These rings act like small prisms, directing light towards a focal point while minimizing lens thickness. Fresnel lenses provide high optical efficiency with minimal distortion, making them a common choice for many VR headsets.

Pancake Lenses utilize folded light paths and polarization techniques to achieve a shorter focal distance, reducing overall headset size. This design enables compact and lightweight XR devices while maintaining image quality. However, due to multiple internal reflections and polarization losses, pancake lenses typically transmit only 20%–30% of the original light intensity, necessitating high-brightness displays to compensate for light loss and maintain image clarity. Additionally, software-based distortion correction is required to mitigate chromatic aberration and ensure a more natural visual experience.

Aspherical Lenses have a non-uniform curvature, unlike traditional spherical lenses. This unique shape reduces optical aberrations and improves image clarity. However, aspherical lenses are more expensive, complex to manufacture, and bulkier, making them less common in consumer-grade XR devices.

3.1.2. Core hardware

Modern mobile devices integrate CPUs and GPUs into a chipset or System on Chip (SoC). For devices tethered to a computer or smartphone, an integrated chipset is unnecessary, as external hardware provides the required computational resources. Most XR devices use the Qualcomm Snapdragon series for their performance and power efficiency. The *chipset* is essential for XR devices, handling network connectivity, spatial tracking, and sensor integration while managing complex graphics and real-time data processing.

However, XR devices face hardware challenges. Rendering immersive 3D environments in real-time requires substantial computing power, leading to issues such as heat dissipation and high power consumption [54]. Maintaining low-latency interactions

and precise spatial tracking demands an optimized chipset [55]. Ensuring high-resolution displays with smooth performance remains a key challenge in XR development.

Tracking systems are crucial to modern VR, directly affecting immersion quality [47]. Tracking identifies a user's viewpoint position and orientation [46]. Additional tracking technologies are discussed in the auxiliary hardware section. There are two main tracking configurations: three degrees of freedom (3DoF) and six degrees of freedom (6DoF). Unlike 3DoF, which tracks only orientation, 6DoF captures positional movement, providing a more immersive experience. Most modern XR devices employ 6DoF for enhanced motion tracking.

Tracking systems are classified as *outside-in* or *inside-out*. Outside-in tracking uses external sensors placed around the play area, but its limited portability makes it less common in modern devices. Inside-out tracking, the more widely adopted method, includes *marker-based* and *camera-based* approaches. In marker-based tracking, transmitters are placed around the environment, while sensors on the headset receive signals, similar to outside-in tracking but in reverse. A common example is lighthouse tracking [56]. In contrast, camera-based tracking relies on integrated cameras and algorithms to determine position. While marker-based systems offer higher accuracy, camera-based tracking is more convenient as it eliminates external hardware. Modern tracking increasingly incorporates AI-powered enhancements, improving response times and tracking accuracy.

3.1.3. Auxiliary hardware

This section introduces auxiliary hardware that enhances interactivity, engagement, and realism in XR environments while not being essential for core functionality.

- **Controllers** are handheld devices that provide an intuitive way for users to interact with virtual environments. They typically feature buttons, joysticks, and motion-tracking sensors.
- **Eye trackers** use single or dual cameras inside the headset to monitor eye movements and predict gaze direction. This technology enables gaze-based interaction and supports foveated rendering.
- **Depth cameras** capture 3D spatial data, improving environment awareness and enhancing object and gesture recognition, especially in MR applications.
- **Motion trackers** detect body movements and are often attached to different body parts to enable full-body motion tracking. They enhance realism in avatars, making them valuable for training and simulation.
- **Haptic suits** provide physical feedback in response to virtual stimuli, simulating touch or force sensations to enhance immersion.

3.2. Software

On top of the hardware layer, various software solutions facilitate the development and accessibility of XR applications. The software ecosystem can be categorized into three main components, as shown in Fig. 7:

- **Operating System**, which abstracts hardware complexity, manages resources, enables real-time processing, and provides essential services for XR applications.
- **Development Tools and SDKs**, including game engines, software development kits, and application programming interfaces (APIs) used for creating and optimizing XR experiences.
- **Human–Machine Interface (HMI)**, which facilitates user interaction with XR environments through controllers, hand tracking, voice commands, and haptic feedback.

3.2.1. Operating system

XR devices rely on specialized operating systems (OS) such as Windows MR, Meta Horizon OS, and visionOS to manage system operations. These OSs ensure efficient execution of XR applications while abstracting low-level operations from the user.

Meta Horizon OS is a highly customized Android-based system designed by Meta for immersive 3D experiences. It optimizes system resources for the Metaverse, supporting VR functionalities such as hand tracking, spatial audio, and high-performance rendering. Similarly, Apple's visionOS is tailored for Apple Vision Pro, facilitating spatial computing with a 3D user interface that seamlessly integrates digital content with the physical world, employing gesture-based interaction, eye tracking, and voice control.

XR OSs must address several core challenges, including real-time resource allocation, latency optimization, and efficient task scheduling [57,58]. These issues are particularly crucial for standalone XR headsets, which must optimize task scheduling and power efficiency in the absence of high-performance computing resources [57,59,60].

OSs typically integrate tracking and sensor algorithms, interfacing with hardware components such as motion, depth, and environmental sensors. These algorithms, often incorporating Simultaneous Localization and Mapping (SLAM) technology, enable XR devices to perform spatial mapping and localization in unknown environments, enhancing immersion and interactivity.

3.2.2. Development tools and SDKs

XR content development relies on game engines and SDKs such as Unity and Unreal Engine for high-performance rendering, OpenXR for cross-platform compatibility, and WebXR for lightweight, browser-based XR applications. Device manufacturers provide specific SDKs, such as Meta SDK and ARKit, to enable XR functionality on their respective platforms. Additionally, APIs like DirectX facilitate XR integration with existing software applications.

Unity is widely adopted in XR development due to its powerful 3D engine and cross-platform support. It features built-in XR integration tools and an extensive plugin ecosystem that simplifies VR/AR application development. OpenXR, developed by the Khronos Group, is an open standard designed to streamline XR application development by reducing the need for multiple codebases.

Many XR development tools integrate AI to enhance their capabilities. For example, ARKit employs AI-based scene understanding and depth estimation to improve AR content interactions. Furthermore, AI-driven approaches enhance rendering efficiency (e.g., deep learning-based super-resolution rendering), optimize gesture recognition (e.g., Transformer-based hand tracking), and improve AR realism through AI-powered lighting and shadow calculations.

3.2.3. Human–machine interface

The HMI in XR encompasses multiple interaction methods, including controllers, hand tracking, and voice interaction. Hand tracking relies on computer vision and AI-driven pose estimation to enable controller-free input, while voice interaction enhances accessibility through AI-powered speech recognition. Additionally, haptic feedback technology simulates tactile sensations, enriching XR experiences with a heightened sense of immersion.

XR user interfaces (UI) extend beyond traditional buttons, menus, and icons to include spatial computing-based 3D interfaces, ensuring that digital content naturally integrates with the real world. Rendering algorithms [61,62] play a crucial role in HMI design by handling image processing, 3D rendering, lighting, and shading to deliver seamless visual experiences. Information

Fig. 8. Preschoolers' use of VR [66].

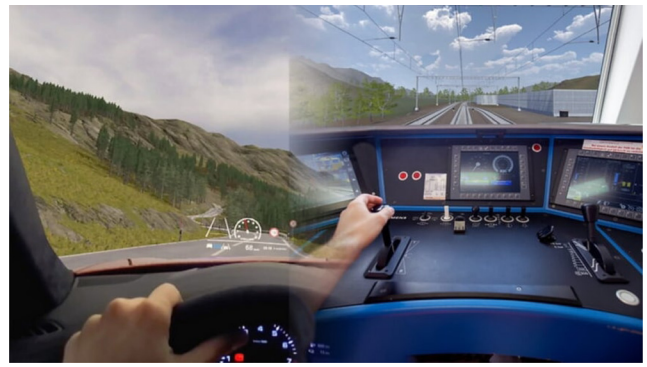

Fig. 9. The Metaverse in driver training [74].

placement [63] is another critical aspect, ensuring that virtual elements such as notifications and data overlays are positioned naturally within the user's FoV.

Haptic terrain rendering [64,65] further enhances immersion by providing tactile feedback that simulates the texture and feel of virtual surfaces. This technology is particularly relevant in training simulations, remote robotic operations, and industrial applications. For example, in medical simulations, haptic feedback allows surgeons to perceive the hardness of tissues, facilitating more precise remote surgical training.

4. Popular domains of metaverse

The Metaverse is already popular in some specific domains, including education, training, medicine, and mixed life. It enables those who are educated to understand knowledge more efficiently, makes training easier and more efficient for trainers, makes medical treatment more flexible and efficient, and makes our lives more colorful. This section will detail and introduce the popular domains of the Metaverse.

4.1. Education

Metaverse is being used in elementary school classrooms to build new classroom models with its rich visualization capabilities. In art classes, Metaverse can bring the art scenes depicted in books to life, making them feel as if they were there. This new classroom model is very popular among children, and compared with traditional classrooms, it shows unique advantages in stimulating students' interest in learning and enhancing the learning experiences [66,67] (see Fig. 8).

In the learning process, as students move up the academic ladder, the content of what they learn will gradually become more complex and abstract, which poses many challenges for teaching. Taking the popularization of radiation knowledge to primary school students as an example, the traditional teaching method faces a dilemma: On the one hand, radiation itself is dangerous, so it is obviously not feasible to let primary school students directly contact the source of radiation to learn the relevant knowledge; on the other hand, relying on the abstract expression to convey the knowledge of radiation is quite difficult for primary school students to understand. The emergence of the Metaverse provides a new way to solve such problems. In the Metaverse's curriculum environment, when conducting radiation knowledge lectures for elementary school students in grades 5 to 6, good interaction between teachers and students is realized, and students are taught to measure the radioactivity of different materials by using simple radiation counters and are guided to observe the effects of distance and shielding materials on the decay of radioactivity. Through this innovative approach, students show a higher level of enthusiasm for learning [68].

Not only in elementary school, but also in high school, the Metaverse plays an active role. Take a high school in Mexico as an example, in the teaching of basic algebra knowledge in mathematics, the abstract mathematical concepts are visualized through AR technology, so that students can learn mathematical concepts and corresponding arithmetic processes in the Metaverse in an interactive and dynamic way. This more vivid and three-dimensional way of teaching achieved remarkable results, with the average scores of the students increasing by 9 percentage points and the error rate in the practice of the problems dramatically decreasing from 18.71% to 2.21% [69].

In today's university education field, the application of the Metaverse is showing increasingly rich and diverse trends. Its influence is not only reflected in the regular classroom teaching, but also extended to the rich and colorful extracurricular activities, which effectively creates a brand-new experience for college students' learning and life, and expands more development possibilities. Many universities have actively engaged in the exploration of interaction with the Metaverse. On the one hand, some universities try to carry out course teaching activities in the Metaverse environment [70,71], breaking the traditional classroom limitations in time and space, and creating an immersive learning atmosphere for students.

On the other hand, many universities have started to establish Metaverse-based laboratories, which, with the advantage of its unique virtual technology, simulate highly realistic practice scenarios for students. This allows them to practice and exercise in a virtual space that simulates a real environment [72].

4.2. Training

The Metaverse is gradually penetrating the traditional training field, bringing significant changes and innovations. Many original training contents are being gradually adapted and transferred to the Metaverse environment.

Take driving skills training as an example, in the traditional mode, automobile, train, and airplane driving training are restricted to specific venues. The Metaverse provides a virtual training space, breaking through the venue constraints, and its immersive experience allows students to have a sense of being in the real scene [73]. Compared to traditional on-site training, this new approach is less costly, more efficient, and safer, as shown in Fig. 9.

In the context of pre-service teacher training, the rapidly repeatable nature of the same task in the Metaverse comes into play. In the virtual classroom environment constructed by Metaverse, pre-service teachers can repeatedly train teaching skills

Fig. 10. Fireman training in Metaverse [76].

Fig. 11. Police training in Metaverse [80].

and optimize teaching methods in front of virtual students, simulating the real teaching process in a low-cost and repeatable way, and effectively improving their teaching ability [75].

Metaverse is not limited by space, and teachers from different regions can share training resources in the same virtual classroom, which helps the professional growth of teachers in remote areas [77].

In the field of workers' pre-service training, Metaverse opens up new paths. Enterprises use it to familiarize new employees with company culture and processes, such as using virtual meeting rooms to promote remote collaboration and cultivate communication and collaboration skills, simulating customer scenarios for service training to enhance customer service awareness and skills [78]. It can also simulate complex industrial scenarios to exercise employees' practical skills [79].

In the field of public safety, Metaverse simulates realistic fire scenarios for firefighters, such as high-rise buildings and other special fires, as shown in Fig. 10, which are used to practice escape and firefighting skills, allowing firefighters to familiarize themselves with different fire scenarios and coping strategies in a safe environment [78].

For police skills training as shown in Fig. 11, it can simulate real-world scenarios such as arresting suspects and rescuing hostages, as well as virtual crime scene search and evidence collection activities to enhance the level of investigation and case handling [78,80].

In conclusion, the Metaverse provides innovative and efficient means and platforms for the training of different professionals, and is gradually changing the pattern and ecology of traditional training.

4.3. Medicine

Metaverse has a wide range of applications in the medical field, which can be used throughout the medical training, psychological treatment, remote treatment, pre-operative analysis, intra-operative guidance, and post-operative rehabilitation, etc., to provide assistance to modern medical treatment. In medical training, the Metaverse enables medical practitioners to practice various operations in safe virtual scenarios by constructing realistic simulation environments to train their professional skills and ability to cope with emergencies. Metaverse can simulate the human skeleton and muscle structure realistically, creating a vivid and intuitive learning experience for students and saving biomaterials, and its 3D perspective helps students understand anatomy deeply [81,82]. In terms of specific skills training, CPR exercises are done in the Metaverse to train standard movements to improve the quality of CPR.

Fig. 12. Surgical remote supervision [93].

Emergency situations, such as simulated choking, emergency intubation exercise intubation skills, and simulated surgical operations to sharpen surgical techniques. Simulation of obstetric surgery, simulation of pregnant women giving birth to practice midwifery skills, and ultrasonography simulation. In conclusion, in the healthcare system, the simulation characteristics of the Metaverse enable all kinds of training skills to be performed in the virtual world [83–87].

The Metaverse plays an active role in both psychotherapy and the popularization of disease concepts. In the field of psychotherapy, its immersive characteristics can be customized according to the patient's needs in order to carry out exposure therapy, so that the patient can face his inner fears and troubles in a safe and controllable virtual situation, which opens up new paths for psychotherapy and helps the patient's recovery [88,89]. In the popularization of the concept of mental illness, VR technology can be used as a tool to popularize the concept of mental illness [88,89]. In the popularization of the concept of mental illness, VR technology can be used as an educational tool for mental illness hallucination, through careful simulation of mental illness audio-visual hallucinations, giving participants a real experience, and improving the public's knowledge of mental illness in a vivid and intuitive way, which is conducive to eliminating related misunderstandings [90].

Metaverse also brings a lot of changes in telemedicine, on the one hand, it supports remote surgical operation, breaks the space limitation with the help of advanced technology, so that the doctor can operate for the patient remotely as shown in Fig. 12, which not only improves the success rate and safety of the operation, but also opens up a new way of optimizing the allocation and sharing of healthcare resources [91]. On the other hand, Metaverse can provide telemedicine services; patients are not bound by geographical location, even if they are in different regions, they can consult with top doctors in a virtual medical room. This advantage is significant in areas where medical personnel are scarce [88,92].

In the preoperative stage, doctors can conduct a more detailed analysis by integrating patient data and 3D models with the help of Metaverse, which provides a strong basis for accurate

Fig. 13. Fitness game [98].

Fig. 14. Tourism in Metaverse [104].

surgical planning. At the same time, the Metaverse can visualize the patient's condition, which not only helps to improve the quality of decision-making and the efficiency of multidisciplinary collaboration, but also allows the patient to know his or her own condition more intuitively [88]. Entering the intraoperative decision-making stage, the Metaverse can provide a more accurate localization view for the primary surgeon, which helps the surgeon to save surgical time and improve the accuracy of surgical localization, thus guaranteeing the smooth and efficient advancement of the surgery.

In the post-operative recovery phase, doctors are always looking for better ways to increase the success rate of surgery and help patients recover faster and better. In this case, Digital Twin in the Metaverse allows doctors to observe and simulate recovery based on a Digital Twin of the patient. In this way, doctors are able to understand the patient's physical state and recovery process in a more intuitive way, and implement preventive treatments for possible complications in advance, thus providing stronger protection and support for the patient's postoperative recovery [94–96].

4.4. Mixed life

The Metaverse is reshaping people's lives, bringing a series of compelling changes and innovative experiences. In the field of health and entertainment, it opens up new paths for traditional exercise, transforming the original fitness activities relying on professional venues into a gamified exercise model in the Metaverse, making the exercise process more interesting and interactive, as shown in Fig. 13 [97].

In terms of shopping, the Metaverse empowers consumers with an unprecedented immersive shopping experience, allowing customers to feel as if they can literally place the product next to them and feel its details and charm. Social gatherings have also been revitalized by the Metaverse, with a variety of virtual social gatherings emerging, such as the world's first virtual entrance ceremony held at a South Korean university [99]. The music field has also benefited from this, with Metaverse concerts flourishing as a new form in which audiences are able to freely interact with singers, cheer with other fans, and enjoy unique musical feasts. In recent years, the number of such concerts has been on a significant upward trend, with the likes of a Metaverse concert of approximately 30 min in duration organized by a virtual music platform [100] and another virtual concert that attracted as many as 10.7 million players, which was broadcast live in real time through avatars, highlighting the powerful appeal of Metaverse concerts [101].

In terms of tourism and cultural exploration, the Metaverse is injecting new energy and opportunities into the tourism industry. People are able to travel to world-famous scenic spots and historical sites in the Metaverse without leaving their homes, and deeply appreciate the customs of different countries, so that they can truly feel the magnificence of nature and the splendor of human civilization without having to travel long distances as shown in Fig. 14 [102,103].

At the same time, it also provides a virtual in-depth exploration of history and culture, allowing users to easily travel back in time to a specific historical period, learn in detail about the process of historical events, and experience the way of life of the people at that time, which greatly enhances the understanding and knowledge of history and culture [102]. By virtue of its unique charm and unlimited development potential, Metaverse is changing people's life style in an all-round way, allowing people to obtain unprecedented diversified experiences and convenient services in the fusion of virtual and reality.

5. Experiences and feedback

The Metaverse, as an emerging digital space, offers immersive experiences that transform entertainment, education, training, and social interactions. However, alongside these opportunities, there are challenges related to hardware, software, usability, and ethical concerns.

This section examines the Metaverse from the user's perspective, highlighting both its benefits and drawbacks. While it enhances immersion, improves learning, and reduces costs, users also encounter hardware limitations, software inefficiencies, physical discomfort, health risks, and ethical dilemmas. By exploring these factors, this section aims to provide a comprehensive understanding of the user experience and identify areas for improvement.

5.1. Positive feedback

The application of the Metaverse in aspects such as immersion, training effect, cost optimization, and safety simulation is significantly enhancing the user experience in fields such as education, training, and medicine. Converging findings from prior studies and user reports indicate broadly positive appraisals of the Metaverse across diverse evaluative dimensions, as shown in Table 1.

5.1.1. Immersion

Across diverse fields, the Metaverse's sense of immersion has garnered high praise. First of all, in terms of simulation experiences, VR is highly acknowledged by practitioners. Because VR can simulate complex situations, enabling trainees to obtain a real

Table 1
User-reported positive experiences with the metaverse.

Aspect	Domain/Context	What improves	Refs
Immersion & Presence	Flight training (PPL); medical education; general education	Sense of presence, decision quality, engagement, motivation, understanding	[105–109]
Training & Learning effect	Corporate VR training; workplace learning	Training efficiency, performance, learner retention, commitment/interest	[110,111]
Teacher & Professional training	Teacher training; emergency medicine skills	Acceptance of training, practical/clinical skills	[77,107]
Driving & Aviation skills	Driver education; PPL theory/preflight/procedures	Skill acquisition, procedural fluency	[73,105]
Subject learning (K–12/HE)	Mathematics education with AR	Academic achievement, classroom attention	[69]
Distance & Maintenance training	Aviation maintenance (remote/distance settings)	Knowledge retention, test performance	[108]
Cost & Safety	Multi-industry training; rail operations; driving practice	Cost/time reduction, safer risk-free practice, flexible access	[73,78,111–113]
Clinical R&D & Surgery	Digital twin for trials; AR-assisted spine surgery	R&D efficiency, surgical accuracy, risk reduction	[91,106]

sense of operation and thereby better master coping skills. For instance, in the training for private pilot licenses, VR simulates various weather conditions, and the created sense of immersion enables trainees to exercise their decision-making abilities more effectively [105]. Second, in the medical field, the Metaverse excels in immersive teaching. The strong sense of immersion brought by the virtual environment provides excellent hands-on practice opportunities for medical students, making them feel as if they were on the spot when conducting virtual dissections and simulated surgical operations [106]. An investigation shows that 88.46% of students are satisfied or very satisfied with the Metaverse immersive teaching. 92.31% of students think it helps deepen the understanding of knowledge points, and 96.15% of students say it has increased their interest in learning, greatly improving the learning effect and student satisfaction [107]. In the field of education, Metaverse education makes learning more vivid and interesting through immersive interactive experiences. In a survey, 88% of students claimed that the Metaverse has increased their interest and participation in learning, and 86% of students said it helps enhance their learning motivation [109]. The results of another survey using the IPQ questionnaire show that the score for spatial presence was 5.08 (out of 6), and the overall presence score was close to 5. These data fully reflect the outstanding performance of the Metaverse in providing a sense of immersion and also demonstrate the high evaluation of the Metaverse's sense of immersion [108].

5.1.2. Training and learning effect

The Metaverse is widely evaluated as having strong advantages and value in training and learning. First of all, the evaluation of the training effect of the Metaverse performs well in many aspects. It greatly shortens the training time. For example, Walmart's early VR training program shows that VR-based training programs can reduce training time by 40%. At the same time, it significantly improves the learning effect. 70% of employees who have received VR training perform better than those who have received training with other materials and technologies, and employee performance improves by 70% [110]. Furthermore, the immersive learning experience of the Metaverse enhances employees' knowledge retention rate and training effect. Surveys show that 78% of learners prefer VR training. It can increase learners' sense of commitment and interest, far exceeding traditional training methods [111]. Second, the Metaverse excels remarkably in specific domains across various fields. In teacher training, a questionnaire survey of 120 teachers showed that more than 54% of the teachers expressed satisfaction with the Metaverse training [77]. In terms of driving training, experiments show that 70.5% of the testers believe that VR driving simulation can effectively help them master the skills required for the driving test [73]. In the training for private pilot licenses, 80% of the respondents believe that VR is very effective in theoretical education, pre-flight inspections, and procedural training [105]. In the training of medical emergency skills, the students in the experimental group performed better than those in the control group in the skills assessment, and achieved significantly higher scores respectively in the assessment items such as the treatment of spinal cord injuries, the treatment of combined injuries, cardiopulmonary resuscitation and tracheal intubation, and the identification of lung segments in bronchoscopy [107]. In mathematics education, after using the AR teaching strategy, the failure rate of students decreased from 18.71% to 2.21%, and the average score increased from 71 to 79 at the same time. Students generally believe that the Metaverse helps them better understand mathematical concepts, and the classroom concentration score was 3.49, showing a significant improvement [69]. In distance education for aviation maintenance, the average score of the learning group using the Metaverse system in the knowledge test was significantly higher than that of the video learning group. In the post-learning test, the average score of the Metaverse system group was 75 points, and that of the video group was 65.5 points. In the knowledge retention test 10 days later, the average score of the Metaverse system group was 68.75, while that of the video group was 53.25 [108]. In short, the Metaverse excels remarkably in training and learning. Whether it is shortening the training time, improving the learning effect, enhancing the knowledge retention rate, or in the specific practices in multiple different fields, it has received good feedback and has been widely recognized and praised.

5.1.3. Cost-effectiveness and safety

The Metaverse demonstrates high cost-effectiveness and safety in multiple fields. Its training content can be reused, significantly enhancing cost-effectiveness. It provides virtual environments for industries such as healthcare, firefighting, policing, military, and high-risk operation scenarios, realistically simulating dangerous and complex operation situations, and effectively reducing the risks of real operations [111,112]. People can conduct training through the Metaverse anytime and anywhere, not being bound by the actual physical location. Even without actual equipment, they can practice independently, effectively saving resources [78]. For instance, VR driving simulation provides drivers with safe practice opportunities, avoids accident risks and property losses, and greatly reduces the time and cost of traditional driving practice [73]. Another study shows that training train and tram drivers using the Metaverse can save about 95% of time and cost compared to traditional methods [113]. The Digital Twin technology of the Metaverse is used in medicine to

Table 2
User-reported negative experiences and challenges with the metaverse.

Aspect	Domain/Context	What suffers/User impact	Refs
Hardware limits	Optics (FOV/brightness), compute demand, battery life	Presence degradation, higher latency, shorter sessions due to power/thermal constraints	[57,62,114–116]
Physical discomfort	VIMS (nausea/dizziness/vertigo), eye strain, posture-related fatigue	Session intolerance; reduced learning/training effectiveness	[48,49,117–120]
Potential health implications	Long-term physical/mental effects; injuries from environmental unawareness	Myopia risk, musculoskeletal pain, headaches; detachment, reduced attention in real tasks; safety risks	[119,121]
Psychological effects	Intense/horror content; avatar embodiment/identity	Anxiety/panic; self-image shifts; identity concerns	[122]
Privacy & Security	Sensitive sensing data (eye/face/body/location); real-world leakage	Privacy harm; surveillance/abuse risks	[123,124]

simulate experiments of drugs and treatment methods, allowing for the observation of treatment effects in advance, accelerating the process of clinical trials, and reducing costs [91]. In actual surgeries, for example, the use of AR technology in spinal surgery to assist in the positioning of pedicle screws is even better than traditional surgical methods, and its accuracy can reach 94.1%. This reduces surgical risks and costs and improves accuracy and efficiency [106]. In conclusion, the Metaverse shows high cost-effectiveness and safety in multiple fields. Its reusable training content, anytime and anywhere training mode, and technological applications reduce costs and risks. In the future, the Metaverse is expected to play an important role in more fields.

5.2. Negative feedback

The Metaverse offers immersive experiences and a new approach to learning and training, but they also present negative effects such as physical discomfort, mental health risks, and privacy concerns due to extensive data collection, as shown in Table 2.

5.2.1. Hardware limitations

Current optical technologies, such as Fresnel or pancake lenses, have various limitations. For example, the limited FoV reduces the immersive quality of virtual environments, and the brightness impacts usability in AR applications. XR devices require significant processing power to render high-resolution and interactive environments [62,114]. AI and real-time tracking also require lots of processing power to fully function [57,115,116]. Current mobile processors struggle with such high demands, which may sacrifice visual quality or create latency to manage power constraints. At the same time, high-powered processors drain batteries quickly, limiting the duration of XR sessions.

Tracking in XR environments is not without its challenges. Even with advanced technology, tracking systems can experience issues such as lag, drift, or complete loss of tracking in dynamic environments. Many sensors struggle in low-light, reflective environments, leading to inconsistent tracking performance. Another limitation in current XR devices is the lack of full-body tracking, reducing the overall effectiveness of the experience. Users may find the disconnection between their real-world movements.

Other hardware limitations impact both usability and user experience, such as feedback device, heat generation, and ergonomics. Current haptic technologies primarily rely on vibrations, which fail to replicate realistic tactile sensations like texture, temperature, or pressure. XR devices can cause noticeable heat buildup when used for a long period of time. Many devices are too heavy and not well-balanced. These factors can lead to physical discomfort, strain, and fatigue, which we may discuss in the next subsection.

5.2.2. Software and user experience issues

Software and user experience are critical factors in the success of XR devices, but they face several challenges. Limited system optimization and inefficient resource management can result in high latency, which disrupts real-time interactions and may cause motion sickness or discomfort.

Many XR platforms lack a robust library of content and have a few content creators. The content cannot move swiftly from one platform to another. Developers face challenges due to inconsistent APIs, limited SDK documentation, or a lack of cross-platform tools. This brings poor user experience and slow innovation, and content creation.

The interaction limitations also affect user experience. Poorly designed UIs can make XR applications difficult to navigate and receive information. For example, the windows placed inside the wall are hard to operate, and the subtitle that is floating around is impossible for user to read. Tracking inaccuracy may cause some trouble in social applications, which will lead to unnatural movement, failure in cooperation, and embarrassment.

5.2.3. Physical discomfort

Physical discomfort is a common concern for XR device users, especially during extended use. The usage of XR devices will lead to motion sickness, eye strain, and fatigue [49,117–119]. Because of the conflict between visual information and the vestibular system, humans may experience visually induced motion sickness (VIMS) [48]. Motion sickness may include nausea, dizziness, vomiting, cold sweats, disorientation, and vertigo [117]. Exposure to screens for a long time in close distance to the eyes can lead to eye strain, headaches, dry eyes, and blurred vision. In XR devices, the fixed focal distance may make this problem more serious [120]. If the weight of HMD is too heavy or with poor weight distribution, it can lead to neck strain or discomfort. XR's advanced haptics and wearable suits can cause localized discomfort or fatigue due to prolonged vibrations or pressure. Some electrotactile suits will cause burns and muscle damage with incorrect settings. XR devices will generate heat during extended use, causing discomfort, particularly in warmer environments or during physically active applications. Extended use in certain postures, such as standing or sitting still while interacting in VR, also leads to muscle fatigue or discomfort.

5.2.4. Potential health implications

Extended use of XR devices can pose potential long-term health risks, both physical and mental, due to their immersive and repetitive nature [121].

Some physical impacts, such as eye fatigue, nearsightedness, musculoskeletal pain, and headaches, will happen similarly to using computers for too long. But it will be more serious due to the distance of the monitor and the fixed focal point. Physical injuries, such as those caused by unawareness of the surrounding environment or improper setup of immersive devices, can result in significant harm.

Apart from physical damage, extended use of XR devices also affects humans' mental health. Highly immersive virtual environments may encourage detachment from reality, potentially leading to social isolation or dependency on virtual spaces. Constant cognitive engagement with virtual environments can lead to mental exhaustion or reduced attention spans in real-world activities. Prolonged use of XR devices can lead to a reduced sense of presence, impaired task performance, reduced user response, and changes in users' awareness of body position or coordination in the real world [119].

Certain XR content, such as intense or horror virtual scenes, can trigger anxiety, panic attacks, or heightened stress, particularly in users sensitive to such stimuli. The use of avatars and the ability to embody different personas or appearances in virtual environments can affect self-image and identity [122]. In social XR applications, negative behaviors such as bullying or harassment can feel more personal and harmful due to the immersive nature of these experiences, potentially causing deeper psychological harm.

5.2.5. Social and ethical concerns

As XR technologies become more immersive and integrated into daily life, their content raises significant social and ethical concerns.

Many XR applications collect sensitive data, including eye movements, facial expressions, body motion, and location, raising concerns about how this information is stored, used, and protected. Users operating applications in the real world may become unaware of their surroundings, potentially leading to unintentional data leaks [123,124].

Social interactions in XR are amplified by immersion. Immersive XR experiences can discourage real-world social interaction, leading to social isolation in the real world. XR content can influence users' decisions, behaviors, and even self-image. It is vulnerable to misuse by malicious actors for manipulation or exploitation. Some negative behaviors, such as bullying or harassment, can be more emotionally impactful. Victims may experience severe emotional distress and cause mental damage.

The usage of XR devices may cause greater inequality. Some people who have access to the technology will have better healthcare, education service, and even have greater employment opportunities. High costs of XR hardware and the infrastructure needed to access the Metaverse exclude low-income individuals and communities. Rural or underserved regions often lack the network connectivity, which cannot accessing the full features of the Metaverse. It is possible that some persons lack education or experience with modern technologies of Metaverse. Language, design prejudices, or content that violates their traditions may cause obstacles for other individuals. This issue will be further elaborated in Section 7.

6. Is the metaverse really coming to fruition?

Although the concept of the Metaverse has attracted widespread attention, when defined in terms of its "realization", namely the full maturity of technology and its widespread adoption integrated into daily life, rather than merely commercial profitability, its much-anticipated era has not yet truly arrived. This situation stems from both deep-seated technical and social factors. The following section provides a detailed exploration of these limiting factors.

6.1. Limitations of hardware equipment

Currently, XR devices have not yet achieved a revolutionary breakthrough in hardware. While there has been some progress compared to devices from several decades ago, there have been no fundamental changes in form factor or interaction methods. For instance, head-mounted XR devices rely entirely on head support for weight distribution, and prolonged use can lead to significant discomfort. This not only limits the widespread adoption of XR but also hinders the expansion of its potential applications.

Moreover, the battery life of XR devices represents a significant limitation. Due to the demands of compute-intensive tasks and high-performance displays, the battery life of these devices often fails to meet the extended usage requirements of users. Regarding interaction methods, current input devices—such as controllers or gesture capture systems—are still less precise and flexible compared to traditional keyboards and touch devices, resulting in inadequate interaction efficiency and accuracy. Consequently, this shortfall in user experience fails to meet expectations and hinders the ability to attract a substantial user base.

The cost factor further hinders the widespread adoption of XR devices. Despite the ongoing introduction of new products by hardware manufacturers, such as the Apple Vision Pro, in an effort to achieve breakthroughs through technological advancements, the high prices of these devices continue to render them usable but not user-friendly. For most users, XR devices are perceived more as novelty than as essential productivity tools.

6.2. Limitations of content creation

The prosperity of the Metaverse relies heavily on a rich and diverse content ecosystem. However, when compared to the volume of content created for traditional PCs and mobile devices, content creation for the Metaverse remains insufficient. Although the number of XR-related 3D scenes and assets has increased in recent years, they still fall short in terms of flexibility, vibrancy, and authenticity, making it challenging to meet users' high expectations for immersive experiences.

The high barrier to content creation is another restrictive factor. Designing 3D scenes and producing assets require professional skills and significant financial investment, making it challenging for ordinary users and small to medium-sized developers to engage in content creation. This limitation further restricts the diversity and scale of available content. Additionally, the issues of interaction fluency and naturalness in user experience urgently need to be addressed. Traditional interaction methods fail to meet the efficiency and accuracy requirements necessary for immersive experiences.

6.3. Security and social issues

As the Metaverse continues to evolve, it introduces a host of new cybersecurity and social ethics challenges. As users' behaviors in the virtual world become increasingly diverse, traditional security measures, such as password-based authentication, are no longer sufficient to meet the demands of XR devices. This new environment necessitates innovative solutions, including authentication methods that leverage biometrics and AI. However, these technologies also pose potential risks of data breaches, heightening users' concerns regarding privacy and security. Malicious behaviors that may arise during virtual social interactions, such as virtual harassment or virtual violence, can adversely affect user experience and mental health. Furthermore, the decentralized nature of the Metaverse may create gaps in regulatory and legal frameworks, complicating social governance.

6.4. Future outlook

Although the current development of the Metaverse faces numerous challenges, its future potential remains highly anticipated. Breakthroughs in hardware, particularly the emergence of XR devices that are lighter, possess longer battery life, and offer more natural interactions, will create new opportunities for the Metaverse's advancement. Equally important, the introduction of a "killer" innovation could serve as a significant catalyst for the Metaverse's evolution toward excellence.

In the realm of content creation, the application of AI can significantly lower the barriers to entry and enhance efficiency. For example, through the use of Generative Adversarial Networks (GANs) or large language models, users can more easily create vivid 3D scenes and interactive content. Furthermore, the integration of quantum computing with AI may offer a more secure and efficient network environment for the Metaverse, thereby increasing users' trust in the platform.

Meanwhile, solutions to social issues must be further explored. For instance, formulating clear guidelines for virtual behavior and integrating them with intelligent algorithmic monitoring can help reduce malicious activities in the virtual world. Additionally, new laws, regulations, and social governance models will ensure the long-term, healthy development of the Metaverse.

7. Possible solutions

As the Metaverse continues to evolve, addressing its challenges will necessitate innovative technological advancements. This section explores potential solutions, emphasizing AI and quantum technologies, as well as social problem-solving strategies, to enhance content creation, user experience, security, and computational efficiency.

7.1. AI-enhanced metaverse

7.1.1. Drive content creation

In the construction and development of the Metaverse, all aspects are closely interconnected and mutually reinforcing, with AI technology playing a central role in enhancing and perfecting content creation.

In the realm of AI-generated art [125], neural networks are employed to produce visual content based on user prompts. One notable example is Deep Dream [126], a tool capable of creating psychedelic-style artworks. This innovation has fostered a more diverse creative ecosystem, enabling a greater number of users to engage in and contribute to the creation of digital art. This offers a wealth of diverse artistic materials for creating Metaverse content, along with essential tools for developing future content. However, art materials alone are insufficient; the Metaverse must be enhanced with 3D scenes. Fully automated 3D reconstruction techniques offer high-precision scaling solutions within the 3D reconstruction process. The implementation of these techniques is poised to significantly influence the future development of the Metaverse.

With a solid scene and material foundation, AI also offers support in the areas of 3D content generation and game design assistance. This process streamlines the generation of 3D content by optimizing game mechanics, level design, 3D asset creation, and NPC programming. Tools such as NVIDIA Ominiverse [127], Midjourney [128], Scenario [129], DallE [130], and Stable Diffusion [131] empower developers to efficiently create unique items, magical potions, and personalized NPCs using simple text prompts. This approach reduces reliance on manual modeling and outsourcing while enhancing flexibility and efficiency in game design.

However, static 3D scenes and characters still lack dynamism and realism. Deep learning algorithms, motion capture technology, and neural networks can be utilized to create realistic movements, expressions, and interactions for NPCs in the Metaverse. This automated and enhanced animation process renders virtual character behavior more dynamic and natural, resulting in a more immersive and believable interactive experience for users [132].

From the perspective of the overall dynamic evolution of the Metaverse, AI facilitates its dynamic development, allowing for personalized experiences that are tailored to the unique characteristics of each user. AI creates immersive virtual environments within the Metaverse by generating 3D worlds (e.g., cities [133], objects [134], and ecosystems [135]), as well as virtual character models, imagery, voice, video, and other elements.

This personalized experience encompasses previous advancements in AI-generated art, 3D content generation, game design assistance, AI-assisted 3D animation, and AI applications in the 3D reconstruction process. It enables the entire Metaverse content creation to be tailored to the user's actions and interests, supporting various scenarios such as video games, social interactions, and educational content [136].

7.1.2. Optimize user experience

During the construction and expansion of the Metaverse, AI serves as a crucial driver for enhancing the user experience. This enhancement encompasses various aspects, including avatar development, multi-sensory experiences, virtual environment generation, and the optimization of user interactions [137].

First, the integration of AI technology enhances the realism and vibrancy of virtualized bodies' performance. Through real-time AI algorithms, the avatar presents facial expressions and body language more naturally and smoothly, incorporating techniques such as Metamotivo [138], StyleGAN [139], and Rodin [140]. This enables users to experience heightened emotional feedback and a stronger sense of identity in their interactions with avatars [132].

Secondly, AI can thoroughly analyze users' behavioral patterns, trends, and preferences. It can dynamically adjust the visual, auditory, and tactile elements within the virtual environment and facilitate the real-time generation of virtual spaces and responses, as demonstrated by GeoAI [141], Runway [142], and Sora [143]. Based on user interactions, AI can generate new regional scenes, alter weather conditions, and initiate a series of contextual events in real time [144]. It can also create contextualized auditory environments and simulate tactile feedback, such as metaSparsh, Digit360, and metaDigitPlexus [145], to provide a more realistic interactive experience [146].

In addition, the integration of Natural Language Processing (NLP) technology provides an additional advantage. With a profound understanding of semantics, AI can swiftly identify user needs and provide accurate, personalized responses [136]. This interweaving and integration of multiple senses significantly enhances the depth and breadth of the Metaverse. With the assistance of these technological tools, users in the Metaverse can not only unlock previously unknown virtual spaces but also immerse themselves in diverse and dynamic environments that resemble real ecosystems. This experience fosters a continual sense of novelty and a desire to explore further.

In addition, by predicting user behavior and needs, AI can proactively adjust system responses, thereby minimizing delays and lag during the interaction process. Whether it involves the avatar's action responses, the real-time switching of auditory and visual elements, or the feedback adjustments within the dynamic virtual space, AI can execute these tasks quickly and efficiently. This capability ensures a natural and seamless operational feedback experience within the Metaverse [136]. Such a seamless experience enables users to engage with various functions in the Metaverse through an efficient, continuous, and natural interaction at all times.

7.1.3. Provide security guarantee

Metaverse has attracted numerous users due to its immersive and interactive nature, but it faces significant cybersecurity challenges. Threats such as deep counterfeiting, identity theft, cyberattacks, and fraud are continually emerging, making them difficult to prevent with traditional security methods [11]. AI is efficient, adaptable, and cost-effective, playing a crucial role in cybersecurity protection for the Metaverse. It can analyze vast amounts of data, identify network threats, and optimize network structures to enhance the security capabilities of the Metaverse [147].

In the realm of identity authentication, AI technologies, including deep forgery detection, biometrics, and continuous authentication, offer effective solutions. Deep forgery detection employs deep learning models to analyze image, video, and audio data in order to identify content that has been artificially generated by AI. Biometric identification integrates multiple biometric features through various authentication methods, such as blink rhythm-based systems (e.g., Blinkey [148]), lip movement-based speech recognition systems [149], and gait-based authentication systems (e.g., GaitLock [150]). This multimodal approach enhances the accuracy and security of identity verification [10]. Continuous authentication continuously monitors user behavior and physiological characteristics in real time, facilitating the timely detection of abnormal behavior and the prevention of identity theft [151].

At the same time, the Metaverse stores a substantial amount of user data, including personal information, transaction records, and virtual assets. This data is vulnerable to leakage and tampering attacks, putting users' personal information, privacy, and virtual identities at risk. Additionally, immersive attacks and other techniques may pose further harm to users [152]. AI models such as Darktrace [153], CrowdStrike Falcon [154], and Vectra AI [155] can perform anomaly detection and analyze network traffic and system logs to identify unusual behavior and issue timely alerts [156].

In the virtual economy, AI can automate transaction execution and assess transaction risks using smart contracts, thereby ensuring the security and reliability of these transactions [157].

7.2. Quantum-enhanced XR

Quantum technology has significant potential to improve XR by addressing some challenges such as recognition, scenario generation, user interaction, sensors, security, privacy, connectivity, and computational complexity [158]. Section 5 has demonstrated the significance of the aforementioned challenges.

Quantum technology introduces a way for the evolution of the Internet that is reliable, low-latency, and secure. Through Quantum Key Distribution (QKD) and Quantum Random Number Generators (QRNG), it can ensure secure communication between XR devices, protect sensitive user data, and provide truly random keys for encryption. This provides unparalleled security and privacy with systems that are theoretically unhackable. Beyond security, quantum technology addresses computational bottlenecks by accelerating resource-intensive tasks such as complex rendering and enabling faster, more detailed simulations for Digital Twin and spatial computing. Challenges such as pathfinding or large-scale user interaction mapping will be optimized through quantum algorithms and no longer be a threat in XR. Moreover, Quantum Machine Learning (QML) can enhance XR systems by enabling efficient large-scale data processing for training AI models. It optimizes transmission protocols by predicting data bottlenecks and adapting data flows for smooth performance. Applications leveraging AI—such as adaptive XR environments or real-time user interaction analysis—will see even greater benefits from this integration.

Kwon et al. propose a quantum approach for XR. They employ hybrid quantum kernels to apply an optimized linear statistical method to the complexity, heterogeneity, and interactive nature of XR data and fed the results to a classical support vector machine for efficient computation [124]. Khalid et al. not only illustrate an anonymous quantum semantic communication framework to improve data transmission but also provides a fantastic technology mapping chart that highlights the potential of leveraging quantum advantage for XR [159] .

7.3. Social problem solutions

Inequality in the context of the Metaverse refers to unfair access or opportunities due to economic, geographic, or social barriers, while the Digital Divide highlights the information gap between those with access to technology and those without it. These challenges can limit the Metaverse's inclusivity and widespread adoption.

Social welfare, technological innovation, infrastructure development, and business responsibility may be the answers to these issues. Governments and organizations could provide subsidies, grants for XR hardware and broadband access and even build public XR libraries and labs to offer access to Metaverse. Organizations and schools may start some training programs to improve knowledge of modern technologies of Metaverse and make XR tools more available to a variety of populations. Governments and Businesses may spend money to expand network access to rural and underserved areas. Establish collaborations between private and public sectors to ensure equitable technological distribution.

Designers could take cultural differences into consideration, and create multilingual, culturally sensitive, and accessible XR content. Encourage companies operating inside the Metaverse ecosystem to use universal design principles, inclusive practices, and fair pricing.

8. Conclusion

This study systematically evaluates the current state of the Metaverse, incorporating case analyses and user feedback to explore its core concepts, key technologies, application domains, and existing challenges. The findings indicate that despite significant advancements in computational architectures, AI, XR interaction technologies, and privacy security in recent years, the Metaverse still faces multiple technological and application-related bottlenecks. These include hardware computational limitations, an underdeveloped ecosystem of high-quality content, insufficient data security and privacy protection mechanisms, and suboptimal immersive user experiences. Additionally, existing research primarily focuses on isolated technological modules or specific application scenarios, lacking a comprehensive assessment of the Metaverse ecosystem. This study bridges this research gap by conducting an integrated analysis of technological progress, industry applications, and user experiences, offering a systematic perspective on the future evolution of the Metaverse.

Looking ahead, the sustainable development of the Metaverse necessitates a coordinated advancement in technological innovation, content ecosystem enhancement, security framework reinforcement, and user experience optimization. AI-driven generative content (AIGC), quantum computing-augmented real-time rendering capabilities, and the integration of 5G/6G networks with edge computing are expected to further improve computational performance and interaction efficiency. Concurrently, the deep integration of multiple industries and the evolution of user-generated content (UGC) and professionally-generated content (PGC) ecosystems will contribute to a more diverse and sustainable digital content landscape. Furthermore, establishing robust

identity authentication and privacy protection mechanisms is critical to enhancing security, while optimizing XR device interaction models and immersive experience design will significantly impact user adoption and engagement. Beyond technological and industrial advancements, the governance framework of the Metaverse must also be developed in parallel. As the Metaverse permeates various sectors, including the economy, education, and healthcare, the formulation of regulations concerning digital identity management, data sovereignty, virtual economic governance, and ethical and legal frameworks will become imperative. Future research should further investigate the Metaverse's implications for digital social norms and global regulatory alignment.

In summary, the development trajectory of the Metaverse suggests that its future evolution will be driven by technological advancements, industry convergence, user experience optimization, and the refinement of digital governance frameworks. Despite persisting challenges, its potential impact on intelligent systems, immersive interactions, and digital society construction holds significant research value. The conclusions drawn from this study not only provide theoretical support for further research in XR interaction, AI-driven content generation, and cybersecurity but also offer practical insights for the industry in Metaverse ecosystem development, privacy protection strategies, and user experience enhancement. Additionally, the advancement of the Metaverse requires not only breakthroughs in technology and industry applications but also a synergistic approach to regulatory oversight, social ethics, and data sovereignty governance to ensure its transition from concept to widespread, sustainable implementation. Therefore, we conclude that the Metaverse is not yet fully mature, and its realization requires further development.

CRediT authorship contribution statement

Zhiguo Liu: Writing – review & editing, Writing – original draft. **Yan Huang:** Writing – review & editing, Methodology, Funding acquisition. **Junyu Mai:** Writing – review & editing, Writing – original draft. **Wei Li:** Methodology. **Zhipeng Cai:** Conceptualization. **Yingshu Li:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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